

Validation Matrix

Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK

Project Name: Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK

Document Type: Validation Matrix

Environment: Local VirtualBox Cybersecurity Lab

Primary Systems: Windows 10 Lab Endpoint, Ubuntu 22.04 Wazuh Server, Wazuh Dashboard, Sysmon, Windows Event Logs

1. Document Purpose

The purpose of this validation matrix is to document how each custom detection in the **Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK** project was tested, validated, mapped to MITRE ATT&CK, and supported with evidence.

This document acts as a structured record showing the relationship between controlled test activity, telemetry source, Wazuh custom rule ID, detection ID, MITRE ATT&CK technique, alert status, and collected evidence. It demonstrates that the project was not only built, but also tested in a controlled and repeatable way.

The validation matrix is designed to support professional documentation standards used in SOC operations, detection engineering, SIEM validation, and threat-informed defense workflows.

2. Validation Objective

The main objective of this validation matrix is to confirm that each custom Wazuh detection rule correctly triggers when the expected Windows endpoint behavior occurs.

The validation process confirms the following:

The controlled test command was executed successfully on the Windows 10 endpoint. Sysmon, Windows Event Logs, or Windows Defender generated the expected telemetry. The Wazuh Agent forwarded the event to the Wazuh Manager. The Wazuh Manager decoded, correlated,

and matched the event against the correct custom rule. The Wazuh Dashboard displayed the alert with the expected rule ID, severity, description, and MITRE ATT&CK mapping. Evidence was captured in the form of screenshots, alert JSON files, validation notes, and analyst observations.

This matrix also helps identify detections that are fully validated, partially validated, pending validation, noisy, or requiring further tuning.

3. Scope of Validation

This validation matrix applies only to controlled, safe, lab-based testing performed inside the local Windows Detection Engineering Lab.

The validation scope includes:

Windows PowerShell activity, encoded PowerShell command simulation, local Windows user creation, Windows service creation or modification, Windows Defender / EICAR detection, Sysmon telemetry, Windows Security logs, Wazuh custom rules, Wazuh dashboard alerts, MITRE ATT&CK mapping, screenshot evidence, alert JSON evidence, and analyst validation notes.

This validation matrix does not include testing on public IP addresses, unauthorized systems, production endpoints, real malware, credential theft, phishing, ransomware simulation, command-and-control infrastructure, persistence using harmful payloads, or any activity outside the controlled lab environment.

4. Lab Environment Summary

The lab environment consists of a Windows 10 endpoint connected to an Ubuntu 22.04 Wazuh server through a VirtualBox host-only lab network. The Windows endpoint is configured with Wazuh Agent and Sysmon. The Wazuh server receives endpoint telemetry, applies built-in decoders and custom rules, and displays alerts in the Wazuh Dashboard.

The project uses custom Wazuh rules to detect selected Windows endpoint behaviors. These detections are mapped to MITRE ATT&CK techniques to show how observed activity relates to real-world attacker tactics and techniques.

The validation workflow follows this path:

Controlled test command on Windows endpoint → Windows/Sysmon/Defender telemetry → Wazuh Agent forwarding → Wazuh Manager decoding and rule matching → Wazuh Dashboard alert → Evidence capture → Validation matrix update

5. Systems Involved

System	Role in Validation	Details
Windows 10 Lab Endpoint	Generates controlled test activity	Runs PowerShell, CMD, Sysmon, Wazuh Agent, Windows Event Logs, and Defender
Ubuntu 22.04 Wazuh Server	Receives and processes telemetry	Runs Wazuh Manager, Wazuh Dashboard, indexer, decoders, and custom rules
Wazuh Dashboard	Alert validation and evidence review	Used to verify rule matches, MITRE mapping, timestamps, and alert details
Sysmon	Endpoint telemetry source	Captures process creation, command-line activity, service-related activity, and other endpoint events
Windows Defender	Malware-test telemetry source	Used only with safe EICAR test file validation
VirtualBox Host-Only Network	Lab connectivity	Allows local communication between endpoint and Wazuh server

6. Validation Status Legend

The following status labels are used in this document:

Status	Meaning
Validated	Detection triggered correctly and evidence was captured
Partially Validated	Detection triggered but requires tuning, more evidence, or better MITRE mapping
Pending Validation	Detection has not yet been fully tested
Failed	Detection did not trigger or produced incorrect results

Tuning Required	Detection works but produces excessive false positives or broad matches
Retired	Detection was removed or disabled after testing

7. Evidence Types

The following evidence types should be collected for each detection:

Evidence Type	Description
Screenshot Evidence	Wazuh dashboard alert screenshot, Event Viewer screenshot, or command execution screenshot
Alert JSON	Raw Wazuh alert exported from dashboard or logs
Command Evidence	Command used to generate the controlled activity
Rule Evidence	Custom Wazuh rule ID and detection logic reference
MITRE Evidence	ATT&CK technique ID and tactic mapping
Analyst Notes	Explanation of why the alert matters and how it should be interpreted
False Positive Notes	Known legitimate activity that may trigger the detection
Cleanup Evidence	Proof that temporary test activity was removed, if applicable

8. Custom Detection Inventory

Detection ID	Detection Name	Wazuh Rule ID	MITRE Technique	MITRE Tactic	Validation Status
DET-001	PowerShell Encoded Command Detection	100100	T1059.001 — PowerShell	Execution	Validated

DET-002	Local User Creation Detection	100101	T1136.001 — Local Account	Persistence	Validated
DET-003	Windows Service Creation / Modification	100102	T1543.003 — Windows Service	Persistence / Privilege Escalation	Validated
DET-009	Defender / EICAR Test Detection	100108	T1204.002 — Malicious File	Execution	Validated
DET-010	Additional Windows Detection Rule	100109	Project-specific	Project-specific	Pending / Optional

9. Master Validation Matrix

Detection ID	Test Scenario	Test Command / Action	Telemetry Source	Wazuh Rule ID	MITRE Mapping	Expected Result	Actual Result	Status	Evidence
DET-001	PowerShell encoded command and execution	Encoded PowerShell command executed from Windows endpoint	Sysmon Event ID 1 / Windows PowerShell logs	100100	T1059.001 — PowerShell	Wazuh generates alert for encoded PowerShell execution	Alert generated in Wazuh dashboard	Validated	powershell-encoded-command-alert.json, screenshot
DET-002	Local user account creation	<code>net user labuser P@ssw0rd123 /add</code>	Windows Security Logs / Sysmon Proc	100101	T1136.001 — Create Account: Local	Wazuh generates alert for local user creation	Alert generated in Wazuh dashboard	Validated	local-user-created-alert.json, screenshot

			ess Creat ion		Acco unt				
DET -003	Windo ws servic e creati on or modifi cation	sc.exe create LabService binPath= "C:\Windows\ System32\cmd .exe /c whoami"	Sysm on Event ID 1 / Wind ows Syste m Logs	10 01 02	T154 3.003 — Wind ows Servi ce	Wazuh generates alert for service creation/m odification	Alert gener ated in Wazu h dash board	Vali date d	service-c reation-a lert.json, sreensh ot
DET -009	Defen der detect s EICA R test file	EICAR test file saved on Windows endpoint	Wind ows Defe nder / Secu rity Logs	10 01 08	T120 4.002 — Malici ous File	Defender detects EICAR and Wazuh generates alert	Alert gener ated or pendi ng dash board verifi cation	Vali date d / Part ially Vali date d	Defende r/EICAR sreensh ot, alert JSON
DET -010	Additi onal projec t-spec ific Windo ws behav ior	[Insert command/action]	Sysm on / Wind ows Event Logs	10 01 09	[Inser t MITR E techn ique]	Wazuh generates expected alert	[Inser t result]	Pen ding / Opti onal	[Insert evidence]

10. Detailed Detection Validation Records

DET-001 — PowerShell Encoded Command Detection

Detection Summary

DET-001 validates whether Wazuh can detect suspicious encoded PowerShell command execution from the Windows endpoint. Encoded PowerShell is commonly associated with malicious or suspicious activity because it can be used to hide command content from casual inspection. In this lab, the test is performed safely using a harmless command.

Detection Details

Field	Value
Detection ID	DET-001
Detection Name	PowerShell Encoded Command Detection
Wazuh Rule ID	100100
Technique ID	T1059.001
Technique Name	PowerShell
Tactic	Execution
Data Source	Sysmon Event ID 1, Windows process creation telemetry
Endpoint	Windows 10 Lab Endpoint
Validation Status	Validated

Test Command

Run the following command on the **Windows 10 Lab Endpoint** using PowerShell or CMD:

```
powershell.exe -EncodedCommand dwBoAG8AYQBtAGkA
```

This encoded command represents a harmless **whoami** execution.

Expected Telemetry

The Windows endpoint should generate process creation telemetry showing PowerShell execution with an encoded command argument. Sysmon should capture the process name, command line, parent process, user context, process GUID, and timestamp.

Expected Wazuh Result

Wazuh should generate an alert matching custom rule ID **100100**. The alert should indicate suspicious encoded PowerShell execution and should map to MITRE ATT&CK technique **T1059.001**.

Validation Evidence

Evidence Item	Expected File / Screenshot
Wazuh alert JSON	powershell-encoded-command-alert.json
Wazuh dashboard screenshot	screenshot showing rule 100100
Event Viewer screenshot	Sysmon Event ID 1 showing PowerShell command line
Command screenshot	PowerShell encoded command execution

Analyst Validation Notes

This detection is useful because encoded PowerShell is often used to hide command content. However, this detection should not automatically be treated as high-confidence malicious activity in isolation. Some administrative tools may also use encoded PowerShell. The rule should be reviewed together with parent process, user account, command-line arguments, destination network activity, and frequency of execution.

False Positive Considerations

Possible false positives include administrative scripts, IT automation, software deployment tools, endpoint management scripts, and legitimate PowerShell automation frameworks.

Cleanup Required

No persistent system change is created by this command. No cleanup is required.

DET-002 — Local User Creation Detection

Detection Summary

DET-002 validates whether Wazuh can detect local Windows account creation. Local account creation can be legitimate, but it is also a common post-compromise behavior used by attackers to establish additional access to a system.

Detection Details

Field	Value
Detection ID	DET-002
Detection Name	Local User Creation Detection
Wazuh Rule ID	100101
Technique ID	T1136.001
Technique Name	Create Account: Local Account
Tactic	Persistence
Data Source	Windows Security Logs, Sysmon Process Creation
Endpoint	Windows 10 Lab Endpoint
Validation Status	Validated

Test Command

Run the following command on the **Windows 10 Lab Endpoint** using Command Prompt or PowerShell as Administrator:

```
net user labuser P@ssw0rd123 /add
```

Optional verification command:

```
net user labuser
```

Expected Telemetry

The Windows endpoint should generate evidence of local account creation. Depending on logging configuration, this may appear in Windows Security logs, Sysmon process creation logs, or both. Wazuh should receive this telemetry through the Windows agent.

Expected Wazuh Result

Wazuh should generate an alert matching rule ID **100101**. The alert should identify local user creation behavior and map it to MITRE ATT&CK technique **T1136.001**.

Validation Evidence

Evidence Item	Expected File / Screenshot
Wazuh alert JSON	local-user-created-alert.json
Wazuh dashboard screenshot	screenshot showing rule 100101
Windows command screenshot	<code>net user labuser /add</code> execution
Event Viewer screenshot	Windows Security or Sysmon event related to account creation

Analyst Validation Notes

This detection has strong SOC value because unexpected local account creation may indicate unauthorized persistence or privilege misuse. During investigation, the analyst should verify who created the account, whether the user was added to privileged groups, whether the account was created during a maintenance window, and whether similar activity occurred across other endpoints.

False Positive Considerations

Possible false positives include legitimate administrator activity, IT onboarding, local lab setup, automated provisioning, endpoint repair activity, and temporary test accounts.

Cleanup Required

After validation, remove the test user account:

```
net user labuser /delete
```

Confirm removal:

```
net user labuser
```

Expected result should indicate that the user does not exist.

DET-003 — Windows Service Creation / Modification Detection

Detection Summary

DET-003 validates whether Wazuh can detect Windows service creation or modification. Attackers may create or modify Windows services to execute payloads, maintain persistence, or run commands with elevated privileges.

Detection Details

Field	Value
Detection ID	DET-003
Detection Name	Windows Service Creation / Modification
Wazuh Rule ID	100102
Technique ID	T1543.003
Technique Name	Create or Modify System Process: Windows Service
Tactic	Persistence / Privilege Escalation
Data Source	Sysmon Event ID 1, Windows System Logs
Endpoint	Windows 10 Lab Endpoint
Validation Status	Validated

Test Command

Run the following command on the **Windows 10 Lab Endpoint** using Command Prompt or PowerShell as Administrator:

```
sc.exe create LabService binPath= "C:\Windows\System32\cmd.exe /c whoami"
```

Optional verification command:

```
sc.exe query LabService
```

Expected Telemetry

The endpoint should generate telemetry showing **sc.exe** being used to create a Windows service. Sysmon should capture process creation, command line, parent process, user context, and timestamp. Windows System logs may also record service creation activity.

Expected Wazuh Result

Wazuh should generate an alert matching rule ID **100102**. The alert should identify Windows service creation or modification behavior and map it to MITRE ATT&CK technique **T1543.003**.

Validation Evidence

Evidence Item	Expected File / Screenshot
Wazuh alert JSON	service-creation-alert.json
Wazuh dashboard screenshot	screenshot showing rule 100102
Command screenshot	sc.exe create LabService execution
Event Viewer screenshot	Sysmon Event ID 1 or System service event

Analyst Validation Notes

This detection is valuable because service creation is a common persistence and privilege escalation technique. During analysis, the service name, binary path, parent process, user context, and timestamp should be reviewed carefully. Unknown services pointing to suspicious paths should be treated with higher priority.

False Positive Considerations

Possible false positives include legitimate software installation, driver installation, Windows updates, endpoint security tools, VPN clients, backup agents, and administrative maintenance.

Cleanup Required

After validation, remove the test service:

```
sc.exe delete LabService
```

Confirm removal:

```
sc.exe query LabService
```

Expected result should indicate the service does not exist.

DET-009 — Windows Defender / EICAR Detection

Detection Summary

DET-009 validates whether Wazuh can detect Windows Defender alerts generated by the safe EICAR antivirus test file. EICAR is not real malware. It is a standard harmless test string used to verify antivirus detection and alerting pipelines.

Detection Details

Field	Value
Detection ID	DET-009
Detection Name	Defender / EICAR Detection
Wazuh Rule ID	100108
Technique ID	T1204.002
Technique Name	User Execution: Malicious File
Tactic	Execution
Data Source	Windows Defender logs, Windows Security logs
Endpoint	Windows 10 Lab Endpoint
Validation Status	Validated / Partially Validated

Test Action

Create or download the EICAR test file on the **Windows 10 Lab Endpoint** in a controlled lab environment. Do not use real malware.

If using the EICAR test string, save it as a `.com` or `.txt` file depending on your Defender behavior and lab requirement.

Expected Telemetry

Windows Defender should detect the EICAR test file and generate a Defender-related alert. Wazuh should receive the Defender event through the Windows agent and generate a corresponding alert.

Expected Wazuh Result

Wazuh should generate an alert matching rule ID `100108`. The alert should indicate Defender detection activity and map to MITRE ATT&CK technique `T1204.002`.

Validation Evidence

Evidence Item	Expected File / Screenshot
Defender alert screenshot	Defender detection page
Wazuh dashboard screenshot	screenshot showing rule 100108
Alert JSON	Defender/EICAR alert JSON
Windows Security screenshot	Defender event evidence

Analyst Validation Notes

This detection is useful for validating endpoint protection telemetry flow from Windows Defender to Wazuh. The purpose is not to test malware execution, but to confirm that Defender-generated security events are forwarded, processed, and visible in the SIEM.

False Positive Considerations

Possible false positives may include legitimate antivirus test activity, lab validation, security team testing, and user-downloaded test files.

Cleanup Required

Ensure the EICAR test file is removed or quarantined by Defender. Confirm there is no remaining test file on the endpoint.

DET-010 — Additional Windows Detection Rule

Detection Summary

DET-010 is reserved for an additional Windows detection created as part of the project. This may include suspicious command execution, scheduled task creation, registry Run key modification, network discovery, or another Windows behavior relevant to the project.

Detection Details

Field	Value
Detection ID	DET-010

Detection Name [Insert Detection Name]
Wazuh Rule ID 100109
Technique ID [Insert MITRE Technique]
Technique Name [Insert Technique Name]
Tactic [Insert Tactic]
Data Source Sysmon / Windows Event Logs
Endpoint Windows 10 Lab Endpoint
Validation Status Pending / Optional

Test Command

Insert the exact command used for DET-010:

[Insert command here]

Expected Wazuh Result

Wazuh should generate an alert matching rule ID **100109**. The alert should show the expected rule name, severity, timestamp, agent name, and MITRE ATT&CK mapping.

Validation Evidence

Evidence Item	Expected File / Screenshot
Wazuh alert JSON	[Insert filename]
Wazuh dashboard screenshot	[Insert screenshot name]
Command screenshot	[Insert screenshot name]
Event Viewer screenshot	[Insert screenshot name]

Analyst Validation Notes

DET-010 should only be marked as validated after command execution, telemetry review, Wazuh alert confirmation, MITRE mapping verification, and evidence collection are complete.

11. Evidence Tracking Matrix

Detection ID	Wazuh Rule ID	Required Screenshot	Required JSON Evidence	Required Command Evidence	Evidence Status
DET-001	100100	PowerShell encoded command alert screenshot	powershell-encoded-command-alert.json	Encoded PowerShell command screenshot	Complete
DET-002	100101	Local user creation alert screenshot	local-user-created-alert.json	<code>net user labuser /add</code> screenshot	Complete
DET-003	100102	Service creation alert screenshot	service-creation-alert.json	<code>sc.exe create LabService</code> screenshot	Complete
DET-009	100108	Defender/EICAR alert screenshot	Defender/EICAR alert JSON	EICAR test evidence	Complete / Partial
DET-010	100109	Additional detection screenshot	Additional alert JSON	Command/action screenshot	Pending

12. Command-to-Detection Mapping

Command / Action	Detection ID	Expected Rule ID	Purpose
<code>powershell.exe -EncodedCommand dwBoAG8AYQBtAGkA</code>	DET-001	100100	Simulates encoded PowerShell execution

<code>net user labuser P@ssw0rd123 /add</code>	DET-002	100101	Simulates local account creation
<code>sc.exe create LabService binPath="C:\Windows\System32\cmd.exe /c whoami"</code>	DET-003	100102	Simulates Windows service creation
EICAR test file creation/download	DET-009	100108	Tests Defender-to-Wazuh alert pipeline
[Insert DET-010 command]	DET-010	100109	Tests additional Windows detection behavior

13. MITRE ATT&CK Mapping Matrix

Detection ID	Technique ID	Technique Name	Tactic	Why It Matters
DET-001	T1059.001	PowerShell	Execution	PowerShell is frequently abused for script execution, payload delivery, discovery, and defense evasion
DET-002	T1136.001	Create Account: Local Account	Persistence	Attackers may create local accounts to maintain access
DET-003	T1543.003	Create or Modify System Process: Windows Service	Persistence / Privilege Escalation	Services can be abused to execute commands or maintain persistence
DET-009	T1204.002	User Execution: Malicious File	Execution	Defender/EICAR validation confirms endpoint malware detection telemetry
DET-010	[Insert ID]	[Insert Technique]	[Insert Tactic]	[Insert reason]

14. Validation Workflow

Each detection should be validated using the following workflow:

First, confirm that the Windows endpoint is active in Wazuh. Then execute the controlled test command on the Windows 10 lab endpoint. After that, confirm that the expected Windows Event Log, Sysmon event, or Defender event was generated. Next, confirm that the Wazuh Agent forwarded the event to the Wazuh Manager. Then search the Wazuh Dashboard for the related alert. Open the alert details and verify the rule ID, detection name, severity, agent name, timestamp, and MITRE mapping. Export or capture alert JSON where possible. Take screenshots of the dashboard alert and related Windows-side evidence. Finally, update the validation matrix with status, evidence filename, and analyst notes.

15. Wazuh Dashboard Validation Checklist

For each detection, confirm the following in Wazuh Dashboard:

Validation Check	Yes / No	Notes
Alert appeared in Security Events		
Correct Windows agent name shown		
Correct rule ID shown		
Correct rule description shown		
Correct MITRE technique shown		
Timestamp matches test window		
Source telemetry is correct		
Alert severity is reasonable		
Raw JSON contains expected fields		
Screenshot captured		
JSON evidence saved		

16. Detection Quality Review Matrix

Detection ID	Detection Quality	Confidence	False Positive Risk	Tuning Required	Analyst Notes
DET-001	Strong	Medium	Medium	Yes	Encoded PowerShell is suspicious but may occur in admin automation
DET-002	Strong	High	Low to Medium	Minimal	Local user creation is high-value but may be legitimate admin activity
DET-003	Strong	Medium	Medium	Yes	Service creation is useful but can trigger during software installs
DET-009	Medium to Strong	High	Low	Minimal	EICAR is safe validation but should be documented as lab-only
DET-010	Pending	Pending	Pending	Pending	Requires final test and evidence

17. False Positive Review

A detection should not be considered complete unless false positives are considered.

Detection ID	Possible False Positives	Recommended Tuning
DET-001	Admin scripts, software deployment, remote management tools, PowerShell automation	Add parent process checks, suspicious flags, encoded command patterns, and user context
DET-002	IT onboarding, local troubleshooting, lab setup, admin account creation	Correlate with admin user, time of activity, group membership changes
DET-003	Software installation, drivers, VPN tools, endpoint protection software, Windows updates	Add allowlist for known service names and signed binaries

DET-009	Security team EICAR testing, antivirus validation activity	Identify EICAR-specific test patterns and label as validation activity
DET-010	Depends on detection logic	Add detection-specific false-positive notes

18. Validation Results Summary

Category		Result
Total Custom Detections Planned	5	
Fully Validated Detections	3–4	
Partially Validated Detections	0–1	
Pending Detections	1	
MITRE-Mapped Detections	4–5	
Detections with JSON Evidence	3+	
Detections with Screenshot Evidence	4+	
Detections Requiring Tuning	DET-001, DET-003	
Highest Confidence Detection	DET-002 — Local User Creation	
Most Useful SOC Detection	DET-003 — Windows Service Creation	
Best Telemetry Pipeline Test	DET-009 — Defender / EICAR Detection	

19. Validation Completion Criteria

A detection should be marked as **Validated** only when all of the following conditions are met:

The test command or action was executed in the Windows lab endpoint. The event was generated in Windows telemetry. The Wazuh dashboard displayed the alert. The alert matched the expected custom rule ID. The alert mapped to the correct MITRE ATT&CK technique. The detection behavior matched the intended use case. Screenshot evidence was captured. JSON evidence or raw alert details were saved. False-positive considerations were documented.

Cleanup steps were completed if the test created a temporary user, service, file, or registry entry.

20. Cleanup Validation

Detection ID	Temporary Artifact Created	Cleanup Command	Cleanup Status
DET-001	No persistent artifact	No cleanup required	Complete
DET-002	Local user <code>labuser</code>	<code>net user labuser /delete</code>	Complete / Pending
DET-003	Windows service <code>LabService</code>	<code>sc.exe delete LabService</code>	Complete / Pending
DET-009	EICAR test file	Remove file or allow Defender quarantine	Complete / Pending
DET-010	Depends on test	[Insert cleanup command]	Pending

21. Analyst Notes

The validation process confirms that the Windows Detection Engineering Lab can generate, collect, process, and alert on selected Windows endpoint behaviors. The project demonstrates a full detection engineering lifecycle: controlled test execution, telemetry collection, SIEM rule matching, MITRE ATT&CK mapping, evidence collection, false-positive review, and documentation.

The detections selected for this project are relevant to common SOC workflows because they represent behaviors that analysts frequently investigate: suspicious PowerShell execution, unauthorized account creation, service-based persistence, endpoint protection alerts, and additional Windows behavior selected for custom validation.

The validation matrix also shows that the project includes operational discipline. Instead of only creating custom rules, the project documents how each rule was tested, what evidence supports it, what false positives may occur, and what cleanup is required after testing.

22. Professional Portfolio Summary

This validation matrix strengthens the project by showing that the custom Wazuh detections were tested systematically and mapped to real attack techniques. It demonstrates hands-on experience with Windows endpoint telemetry, Wazuh SIEM/XDR, Sysmon, Windows logs, MITRE ATT&CK, detection validation, and analyst-style evidence handling.

For a cybersecurity portfolio, this document shows more than tool usage. It shows the ability to design detections, validate them safely, document results, assess false positives, and present findings in a professional format.

23. Final Validation Statement

The detections documented in this validation matrix were tested in a controlled Windows lab environment using safe commands and test actions. The results confirm that the Wazuh detection pipeline can identify selected Windows endpoint behaviors and present them in the Wazuh Dashboard with custom rule IDs and MITRE ATT&CK context.